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33. Return on Investments (ROI): Measures profitability relative to Cost.

Example: $ROI = (\text{profit} \div \text{cost})$

34. Clicking Through Rate (CTR): Percentage of people who click on an ad.

Example: 2% of viewers click an ad.

35. Bounce Rate: Percentage of visitors who leave a site quickly.

Example: 60% leave after one page.

36. A/B Testing: Comparing two versions to see which performs better.

Example: Testing two ad designs.

37. Lead Generation: The process of attracting potential customers.

Example: Collecting emails via a sign up form.

38. Marketing Attribution: Identifying which Marketing channel led to a conversion.

Example: Determining whether Instagram or email drove sales.

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25. Customer Retention Rate: Percentage of customers who stay with a business.

Example: 90% of customers remain active.

26. Net promoter Score (NPS): Measures customer loyalty based on likelihood to recommend.

Example: Survey Score from 0-10

27. Average Order Value (AOV): Average amount spent per ~~purchase~~ ^{Purchase}

Example: Customers spend R250 per order.

28. Sales funnel: The journey from awareness to purchase.

Example: Awareness → Interest → Purchase.

29. Marketing & Campaign Analytics

29. Marketing ~~Campaign~~ Campaign: A planned set of marketing activities to achieve a goal.

Example: A Social media ad Campaign.

30. Post-Campaign Analysis: Evaluating Campaign performance after completion.

Example: Measuring ROI after an ad Campaign.

31. Incremental Revenue: Extra revenue generated due to a Campaign.

Example: R50 000 additional sales from ads.

32. Return on Ad Spend (ROAS): Revenue generated per advertising Cost. Example: R5 earned for every R1 spent.

19. Benchmark: A benchmark is a standard used to compare performance.

Example: Comparing your sales to industry averages.

20. Segmentation: Segmentation groups customers based on shared characteristics.

Example: ^{Dividing customers by age group.} ~~Comparing your sales to industry averages.~~

03 Customer & Sales KPIs

21. Conversion Rate: Percentage of users who take a desired action.

Example: 5% of websites visitors make a purchase.

22. Customer Acquisition Cost (CAC): The cost of acquiring one new customer.

Example: Spending R500 to gain one customer.

23. Customer Lifetime Value (CLV): Total revenue a business expects from a customer over time.

Example: A customer spends R10,000 over 5 years.

∴ CLV : CAC Ratio: Compares value earned vs cost to acquire a holding ratio is usually 3:1

Example: $CLV = R900$, $CAC = R300 \rightarrow$ Ratio 3:1

24. Churn Rate: Percentage of customers who stop using a service.

Example: 10% monthly subscription cancellations.

Example: Social Media posts.

12. Structured Data: Structured data organised in rows and columns making it easy to store and analyse.

Example: Excel Spreadsheets.

13. Data warehouse: A data warehouse is a central system that stores large amounts of structured data for analysis.

Example: A company storing years of sales data in one system.

14. Data Mining: Data mining is the process of discovering patterns and relationships in large datasets.

Example: Finding buying trends in customer data.

15. Data Visualisation: Data visualisation presents data in visual formats like charts and graphs.

example: A bar chart showing sales performance.

16. Dashboard: A dashboard is a visual display of key metrics in real time.

Example: A live sales tracking dashboard.

17. Metric: A metric is any measurable value used to track performance.

Example: Website visits.

18. Key Performance Indicator (KPI): A KPI is a critical metric tied directly to business goals. Not all metrics are KPIs.

Example: Revenue growth rate as a KPI.

5. Big data: Big data refers to extremely large and complex data-sets that require advanced tools to process due to their ~~massive~~ volume, speed and variety.

Example: Social media platforms analysing millions of user interactions daily.

6. Descriptive Analytics: Descriptive analytics explains what has happened in the past using historical data.

Example: A report showing last year's revenue.

7. Diagnostic Analytics: Diagnostic analytics explains why something happened by identifying causes and relationships.

Example: Analysing why sales dropped in a specific month.

8. Predictive Analytics: Predictive analytics uses data and models to forecast future outcomes.

Example: Predicting customer demand for next quarter.

9. Prescriptive Analytics: Prescriptive analytics suggests action to achieve desired outcomes based on predictions.

Example: Recommending ^{Pricing} ~~marketing~~ strategies to maximise profit.

10. Data Literacy: Data Literacy is the ability to read, understand and communicate data effectively.

Example: A manager interpreting a sales chart correctly.

02 Data, Measurement & Reporting

11. Unstructured Data: Unstructured data is organised in a pre-defined format such as text and images.

COURSE NAME: Data Analytics

Name: Millicent Jhingan-ga

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THE LANGUAGE OF BUSINESS ANALYTICS

01 FOUNDATIONAL CONCEPTS OF ANALYTICS

1. **Business Analytics:** Business analytics is the process of using data, statistical methods and tools to support business decision making and improve performance. It focuses on solving business problems using data insights.
Example: A retail company analyses ^{Sales} data to decide which products to restock.
2. **Data Analytics:** Data analytics focuses on examining raw data to find patterns, trends and insights. It is broader and not always tied directly to business decisions.
Example: A data analyst studies website traffic patterns without necessarily making strategic decisions.
3. **Business Intelligence (BI):** BI refers to tools and systems used to collect, visualise and report data for decision-making. It mainly focuses on historical and current data.
Example: A dashboard showing monthly sales performance.
4. **Data-driven decision making:** This is the practice of making decision based on data rather than intuition or guesswork.
Example: A company launches a product based on customer demand data.